Matthew Berger

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Senior Staff Rust engineer with 5+ years of production Rust, writing Rust since 2016. Founding software engineer at Hyphen, where I built a food assembly robotics controls suite from zero to production, including a real-time async pub/sub message broker over TCP with cross-host bridging running at sub-millisecond local latency with zero message loss across 3+ years of continuous fleet operation. Background spans embedded firmware to cloud, with prior production experience in aerospace imaging (Sierra Nevada) and safety-critical medical robotics (Hamilton). Author of Rust crates (freecs, enum2 family) with 280k+ combined crates.io downloads.

Work Experience

Hyphen (Culinary Robotics) — Remote
Senior Staff Software Engineer (Founding Engineer) — *January 2026 – May 2026*

- Sole architect of the production food assembly robotics controls suite, with 1,000+ pull requests merged across firmware, controls, IPC, motion, and cloud
- Shipped Hyphen Studio, a Dioxus desktop application enabling operators to configure makeline layouts and parameters directly
- Replaced TypeScript/GraphQL tablet portal with a pure Rust Leptos rewrite connecting directly to the IPC broker, eliminating an entire backend tier

Hyphen (Culinary Robotics) — Remote
Staff Software Engineer (Founding Engineer) — *September 2022 – January 2026*

- Designed and built the production async pub/sub message broker over TCP with cross-host bridging, coordinating every distributed process across the industrial PC and embedded boards, at sub-millisecond local latency and zero message loss across 3+ years of continuous fleet operation
- Built multi-axis motion control with trajectory planning, kinematics, and motor drivers with closed-loop PID for MG4010 and LKMG actuators
- Built the full stack from bare-metal firmware (RP2040/Embassy) to distributed robotics control system, custom Yocto Linux distribution, and AWS cloud
- Built Hyphen Explorer (Bevy + egui) for real-time IPC visualization, debugging, and system control, achieving 100% adoption across all Hyphen engineers
- Mentored team of web developers into Rust systems programmers through architectural review, pair programming, and documented patterns

Hyphen (Culinary Robotics) — Remote
Senior Software Engineer (Founding Engineer) — *July 2021 – September 2022*

- Diagnosed scaling limitations in legacy PLC system preventing growth beyond prototype
- Led technical pivot from costly PLC-based control system to embedded Rust architecture
- Pioneered RP2040 firmware with Embassy-rs, creating first Rust TMC5160 motor and AS5048A encoder drivers proving Rust could meet real-time control loop requirements

Sierra Nevada Corporation — Englewood, CO
Software Engineer III — *May 2020 – July 2021*

- Developed C++/Rust aerospace imaging system capable of collecting and orthorectifying 5GB/sec of pixel data during a flight
- Built asynchronous Rust simulator for unavailable flight hardware, saving 3 months on project timeline

Scientific Games — Reno, NV
Software Engineer — *July 2019 – May 2020*

- Worked on game engine architecture in the GDK (Game Development Kit), Scientific Games' Unity-based engine powering many casino game titles

Hamilton Company Reno, NV
Software Engineer (Intern through Full-Time) *October 2014 – July 2019*

- Built safety-critical software in a cross-disciplinary environment that calibrates and operates liquid-handling medical robots
- Reduced calibration development from 2 months to 2 weeks by consolidating multiple applications into a single GUI and reusable plugin framework
- Created Windows VM bootstrapper that cut development environment setup from a day to a few clicks
- Developed calibration application driving Hamilton's firmware via UI for Illumina, Hamilton's largest OEM customer ($2M in machine purchases)
- Mentored new hires and interns on team processes and coding practices
- Automated gravimetric analysis QA process, saving 1+ week per robot across production fleet
- Saved developers hours per day on common tasks by creating a suite of in-house tools
- Improved instrument sales by creating software adapters for both SiLA and non-SiLA compliant devices through third-party device manufacturer collaboration

Skills

Languages: Rust, C++, C, C#, TypeScript, Python
Systems: Real-time low-latency systems, embedded (RP2040, Embassy), async Rust (Tokio, async-std), IPC, message brokers, networking, Yocto Linux, distributed systems
Cross-Platform: Linux, macOS, Windows, Android, WebAssembly, Steam Deck, OpenXR/VR
UI/Graphics: Leptos, Dioxus, egui, Bevy, wgpu, Vulkan, DX12, Metal, OpenGL
Cloud/Infra: AWS, Pulumi, Greengrass IoT, Docker, GitHub Actions

Open Source

- **Nightshade**: 200,000+ line data-oriented Rust game engine featuring a custom render graph, built-in editor, rapier3d physics, kira audio, gltf asset loading, procedural terrain, and Steam integration. Runs natively on DX12/Metal/Vulkan and in-browser via WebGPU (live demo: matthewberger.dev/nightshade)
- **freecs** (48k+ downloads on crates.io): archetype-based ECS in Rust with zero unsafe code, Rayon-parallel system execution, sparse-set tags, deferred command buffers, and watermark change detection. The entire ECS is generated at compile time via a declarative macro. Design documented in a 3-part series at matthewberger.dev/articles
- **enum2 macro family** (enum2str, enum2contract, enum2egui, enum2pos, enum2repr): derive macros for enum-driven design patterns with 230k+ total downloads on crates.io
- **wgpu-example**: minimal Rust + wgpu + egui template showing how to build graphics that run everywhere, including native desktop, WebGPU and WebGL via WebAssembly, Android, Steam Deck, and OpenXR VR with hand tracking
- **cameras** (new in 2026): cross-platform Rust library for real-time video frame pipelines across Linux, macOS, and Windows, with data-oriented design (explicit format negotiation, push-based frame delivery, zero trait objects). Ships with Dioxus and egui integrations.

Education

University of Nevada, Reno 2017
B.S. Computer Science & Engineering, Minor in Mathematics